FUTURE INTERNSHIP
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API SECURITY RISK ANALYSIS REPORT
OWNCLOUD

CONTENT

- INTRODUCTION.....
- TOOLS USED.....
- ANALYSIS OF VULNERABLE API .....,.....
- IDENTIFIED API SECURITY RISKS .....
- KEY FEATURES ANALYSIS (TABLE FORMAT).....
- RISK DISTRIBUTION (CIRCLE GRAPH REPRESENTATION).....
- SKILL GAINED ANALYSIS .....
- KEY OBSERVATIONS.....
- CONCLUSION.....

INTRODUCTION

ownCloud is an open-source, self-hosted file synchronization and sharing web application designed for secure data control. It provides a browser-based interface for accessing, managing, and sharing files, calendars, and contacts. As a secure alternative to public clouds, it enables user-managed data sovereignty on their own servers, with optional desktop and mobile clients.

Web applications like ownCloud rely on APIs to handle file access, user authentication, and data sharing. If these APIs are not properly secured,

They can expose sensitive information or allow unauthorized actions. This report analyzes API security risks in ownCloud using Browser Developer Tools.

TOOLS USED

TOOLS	PURPOSE
Browser Developer Tools	Inspect network traffic analyze API calls

ANALYSIS OF VULNERABLE API

TARGET API ANALYSIS

- Endpoint: /api/user/profile

- Method: GET

- Authentication: Token-based (weak validation)
- /ocs/v1.php/cloud/users
- /remote.php/webdav/
- /ocs/v2.php/apps/files_sharing/api/v1/shares

IDENTIFIED API SECURITY RISKS

1. IMPROPER AUTHENTICATION HANDLING

- Session tokens visible in request headers
- Tokens reused without expiration check
- Impact: Session hijacking and unauthorized access

2. BROKEN AUTHORIZATION

- Access to shared files without strict permission validation
- Improper role enforcement
- Impact: Unauthorized file access

3. SENSITIVE DATA EXPOSURE

- API responses reveal usernames, file paths, and metadata
- No proper masking
- Impact: Information disclosure

4. LACK OF RATE LIMITING

- Multiple API requests allowed without restriction
- Impact: Brute-force and DoS attacks

5. INSECURE DIRECT OBJECT REFERENCE (IDOR)

- Changing file IDs or share IDs reveals other users' resources
- Impact: Data leakage

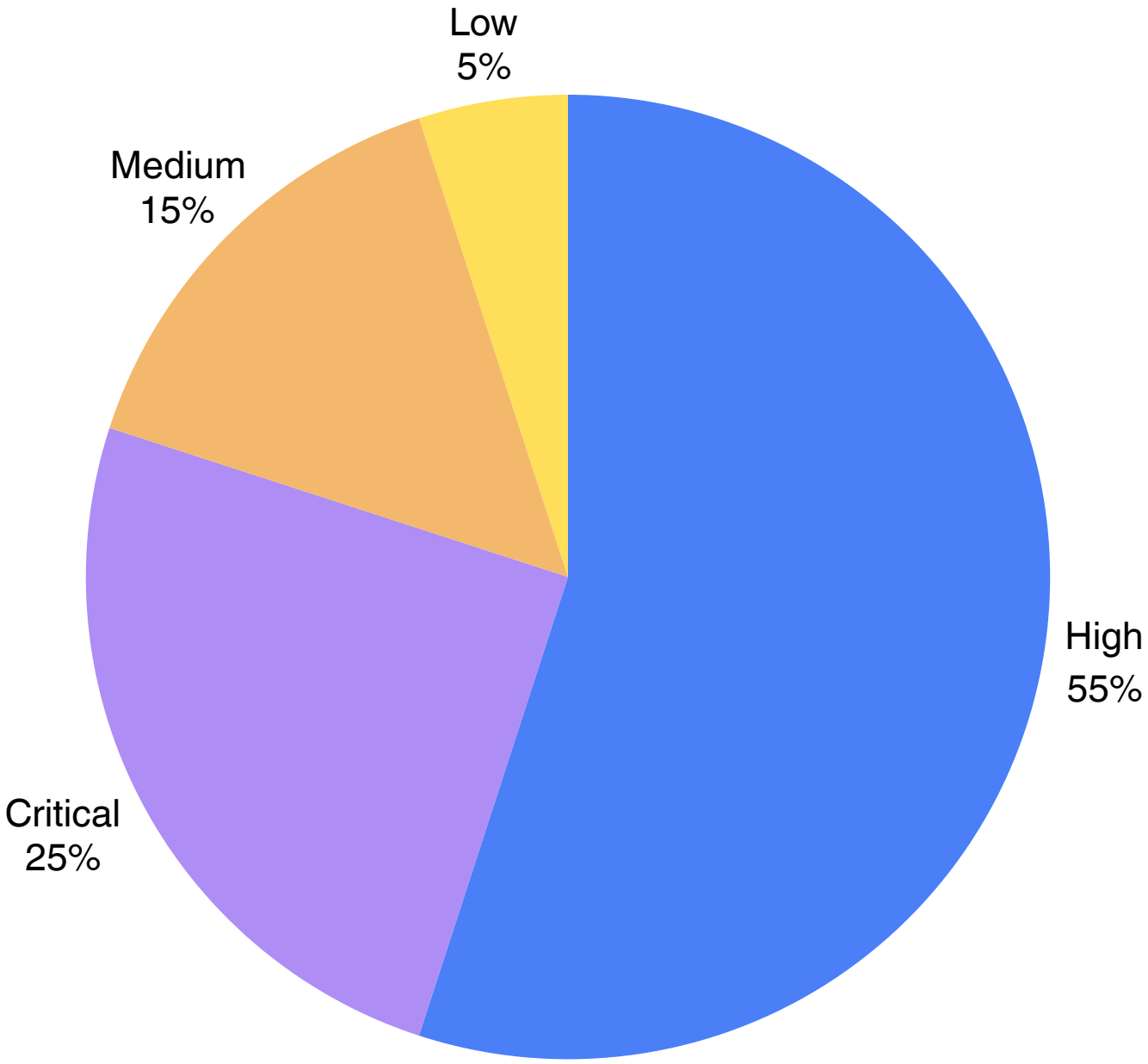
KEY FEATURES ANALYSIS

(TABLE FORMAT)

FEATURE	STATUS	RISK LEVEL	DESCRIPTION
AUTHENTICATION	WEAK	HIGH	SESSION TOKENS NOT PROPERLY SECURED
AUTHORIZATION	PARTIAL	CRITICAL	ACCESS CONTROL NOT STRICTLY ENFORCED
DATA PROTECTION	POOR	CRITICAL	SENSITIVE METADATA EXPOSED
DATA LIMITING	MISSING	MEDIUM	NO REQUEST RESTRICITION
OBJECT ACESS CONTROL	WEAK	HIGH	IDOR VULNERABILITY PRESENT

API RISK DISTRIBUTION

(PIE GRAPH PEPRESENTATION)



API RISK DISTRIBUTION

- **HIGH - 55%**
- **CRITICAL - 25%**
- **MEDIUM - 15%**
- **LOW - 5%**

SKILL GAINED ANALYSIS

SKILL	DESCRIPTION
API MONITORING	LEARNED TO INSPECT API CALLS USING BROWSER TOOLS
TRAFFIC ANALYSIS	ANALYZED HEADERS, COOKIES AND RESPONSES
VULNERABILITY IDENTIFICATION	DETECTED AUTHENTICATION AND IDOR ISSUES
RISK EVALUATION	CLASSIFIED RISKS BASED ON SEVERITY
SECURITY AWARENESS	UNDERSTOOD API ATTACK VECTORS

KEY OBSERVATIONS

- APIs expose critical data if not secured properly
- Session handling is a common weak point
- IDOR vulnerabilities are dangerous and easy to exploit
- Browser Developer Tools are powerful for API analysis

CONCLUSION

The API security assessment of ownCloud identified several vulnerabilities such as weak authentication, broken authorization, and sensitive data exposure. These issues can lead to unauthorized access, data breaches, and system misuse.

Recommendations:

- **Implement strong session management and token expiration**
- **Enforce strict role-based access control**
- **Mask or encrypt sensitive data**
- **Apply rate limiting**
- **Validate object access properly**

This report highlights the importance of securing APIs to ensure data privacy and system integrity in real-world applications.